

# Persistent Homology of Music Network with Three Different Distances

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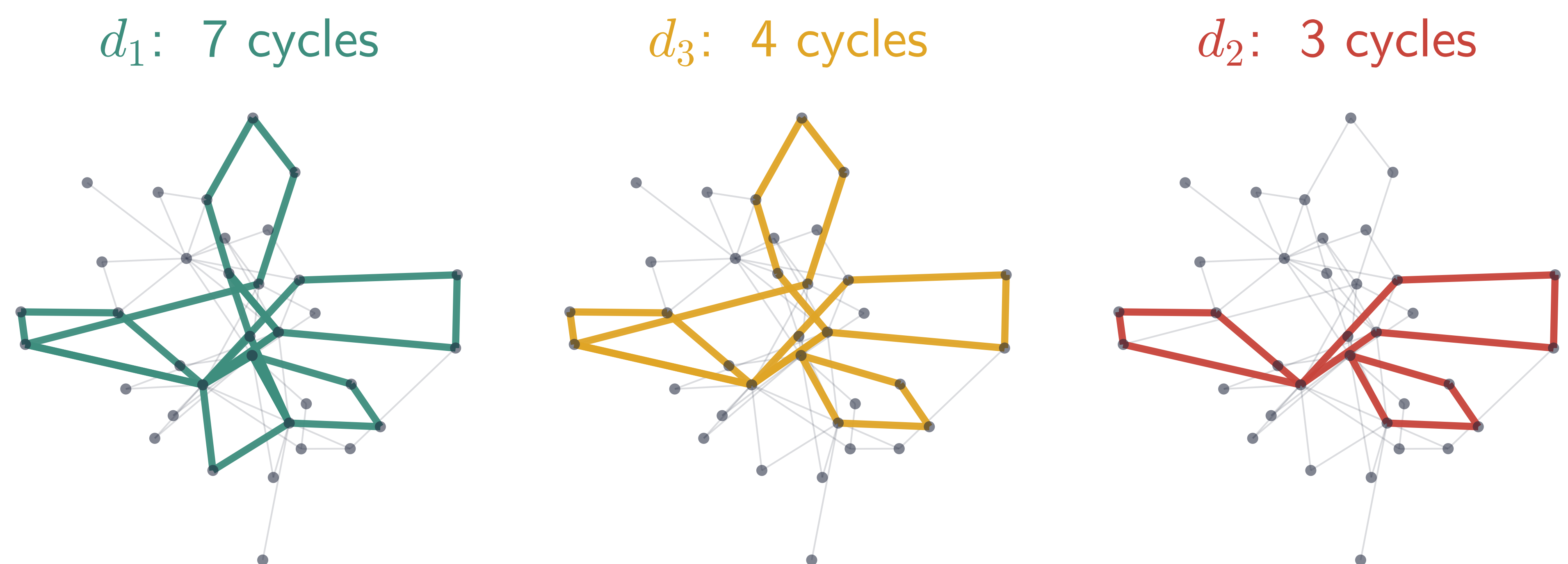
Project page

## The big idea

Distance is a **tunable resolution knob** on musical structure — sweeping  $d_1 \rightarrow d_3 \rightarrow d_2$  distills a piece down to its **core cyclic skeleton**.

- **Three path distances:**  $d_1$  (min edges),  $d_2$  (min weight — the only metric),  $d_3$  (hybrid).
- **Proven ordering**  $d_2 \leq d_3 \leq d_1$  lifts to a **nontrivial barcode injection**  $\text{bcd}_1(d_2) \hookrightarrow \text{bcd}_1(d_3) \hookrightarrow \text{bcd}_1(d_1)$  (Heo–Choi–Jung 2025).
- **Validated** on real Korean court music — an exact match to the paper’s Table 1.

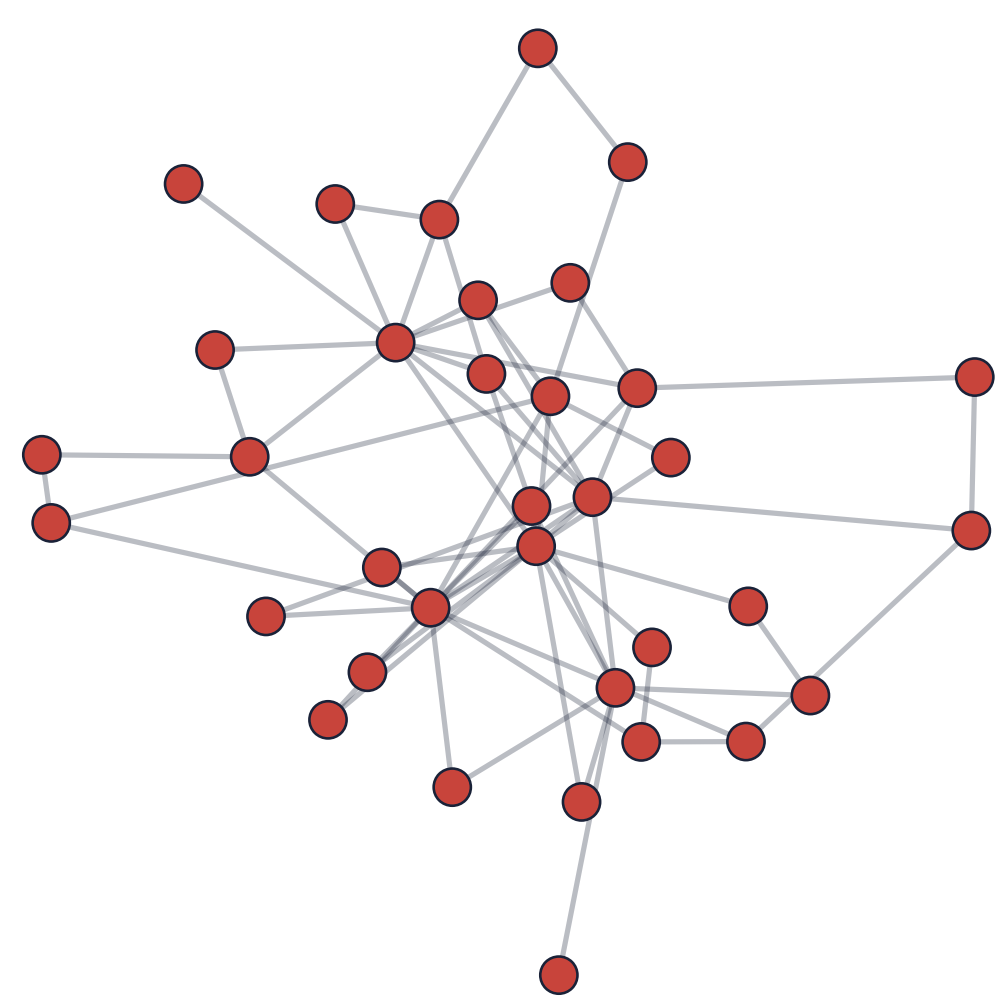
## Cycle skeleton extraction



On *Sanghyeondodeuri* (geomungo): as  $d_1 \rightarrow d_3 \rightarrow d_2$ , the  $H_1$  barcode thins  $|\text{bcd}_1| = 7 \rightarrow 4 \rightarrow 3$  — only the robust core loops survive.

## From music to a graph

Each **node** = (pitch, duration); an **edge** joins consecutive notes; the **weight** is  $1/(\text{co-occurrence count})$  — frequent neighbors are closer. Rests are skipped.



The music network of *Sanghyeondodeuri* (geomungo).

## A universal ordering

For every pair of nodes,

$$d_2(v, w) \leq d_3(v, w) \leq d_1(v, w).$$

This carries over to one-dimensional persistent homology.

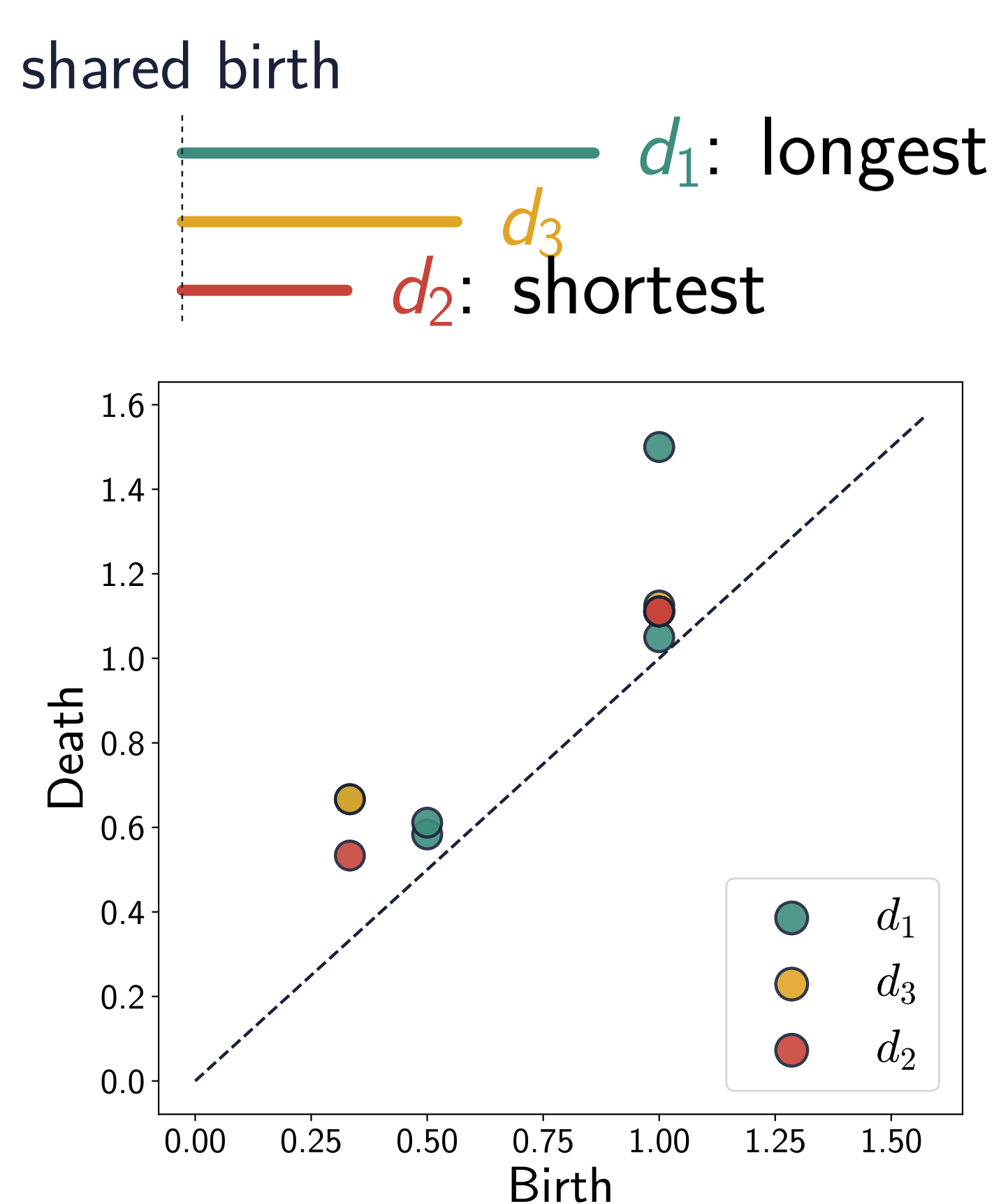
## Barcode injection

The birth edges nest,  $B_2 \subseteq B_3 \subseteq B_1$ , giving injections

$$\text{bcd}_1(d_2) \hookrightarrow \text{bcd}_1(d_3) \hookrightarrow \text{bcd}_1(d_1).$$

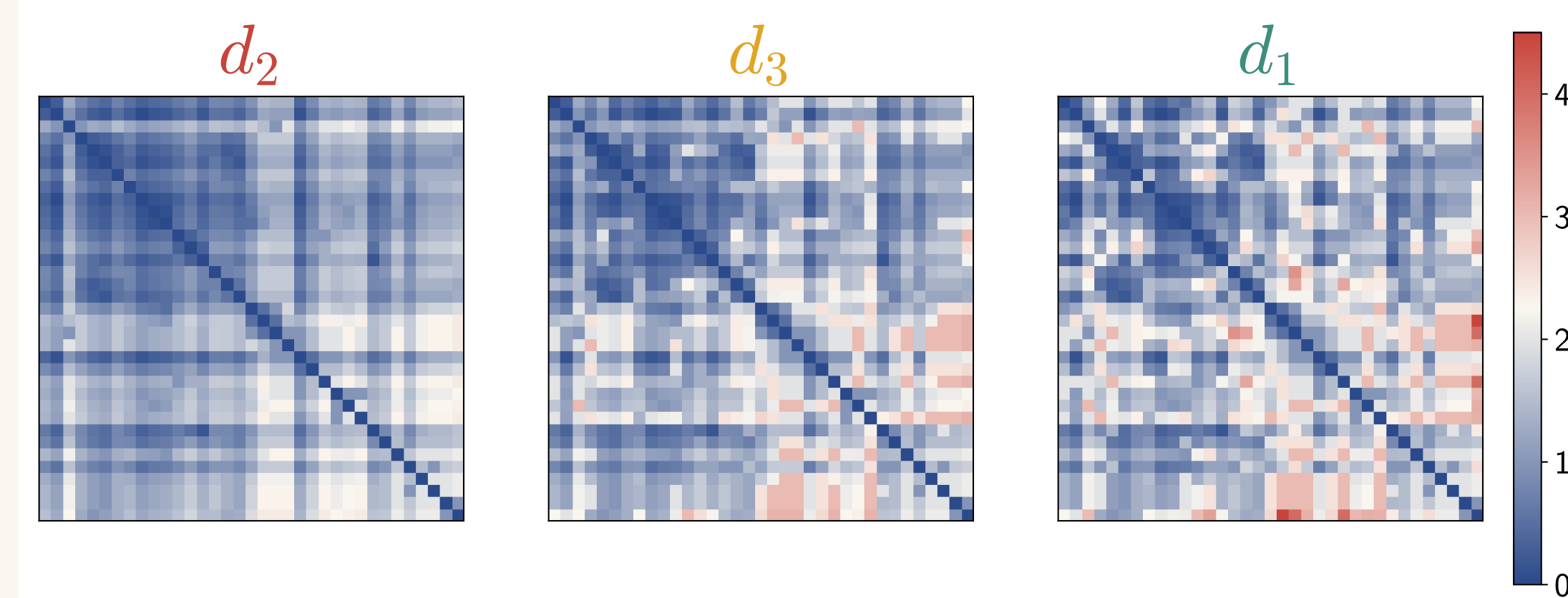
A **nontrivial theorem** for path-representable distances (Heo–Choi–Jung 2025, arXiv:2501.03553).

Each cycle keeps its **birth**; its **lifespan** grows  $d_2 \leq d_3 \leq d_1$ , and the number of cycles can only increase.

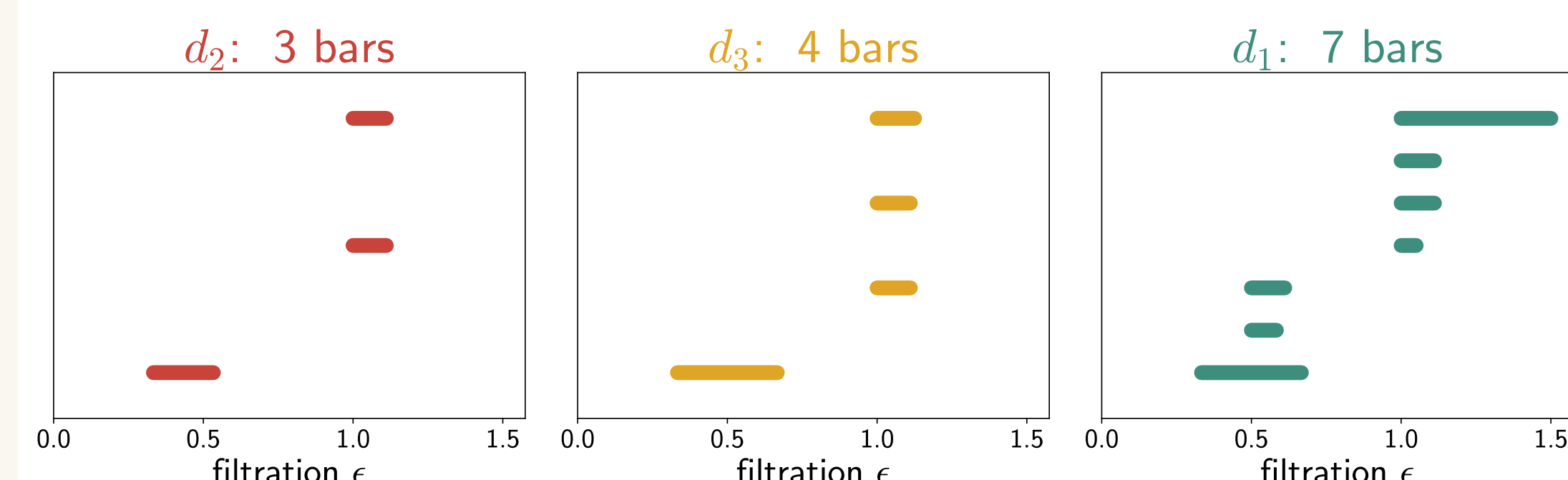


On real music:  $d_1$  points lie above  $d_2$  — longer lifespans, shared births.

## Evidence on real music



Distance matrices:  $d_2$  (cool/smooth)  $\rightarrow d_1$  (hot) the ordering is visible.



$H_1$  barcodes:  $|\text{bcd}_1(d_1)|, |\text{bcd}_1(d_3)|, |\text{bcd}_1(d_2)| = 7, 4, 3$ ; lifespans grow, births shared.

## Distilling the core

Ranking  $d_1$  cycles by how far down they survive ( $\rightarrow d_3 \rightarrow d_2$ ) gives a principled **importance hierarchy of musical loops** — a multi-resolution descriptor of a piece.

## Takeaways & future work

- Distance choice = a tunable lens; results are nested, not arbitrary.
- Toward **AI music composition** with multiple stylistic perspectives from one source.



Interactive explorer & audio demos on the project page.

## Only $d_2$ is a metric

$d_2$  satisfies the triangle inequality;  $d_1$  and  $d_3$  can violate it (a heavy direct edge). Yet all three give meaningful musical structure.

## Why it matters

The same piece yields *different but nested* topological readings.  $d_2$  exposes the robust core;  $d_1$  adds finer, peripheral loops. The changes are **systematic, not random**.

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Key refs: Heo–Choi–Jung (2025) arXiv:2501.03553 (barcode injection); Tran–Park–Jung (2023) *J. Math. Music*; Sethares–Budney (2014).

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